MFCRT Based Pitch Detection in Noisy Speech Signals

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Abstract:

This project explains pitch detection using the MFCRT method. It works better in noisy signals and gives accurate results. The method uses correlation and Radon transform to detect pitch.

Introduction:

Pitch detection is used to find the fundamental frequency of a speech signal. It is important in speech recognition and audio processing. In real-world signals, noise makes pitch detection difficult. Traditional methods fail in noisy conditions. Therefore, MFCRT method is used because it provides better accuracy and stability.

Method:

The MFCRT method works in several steps. First, the speech signal is taken as input. Then a pre-emphasis filter is applied to improve the signal quality. The signal is divided into small frames. After that, correlation is calculated across multiple frames. Then Radon transform is applied to detect patterns. Finally, the pitch is estimated from the detected peaks.

Pseudocode:

Start

Input speech signal

Apply pre-emphasis

Divide into frames

Compute multi-frame correlation

Apply Radon transform

Detect peaks

Estimate pitch

End

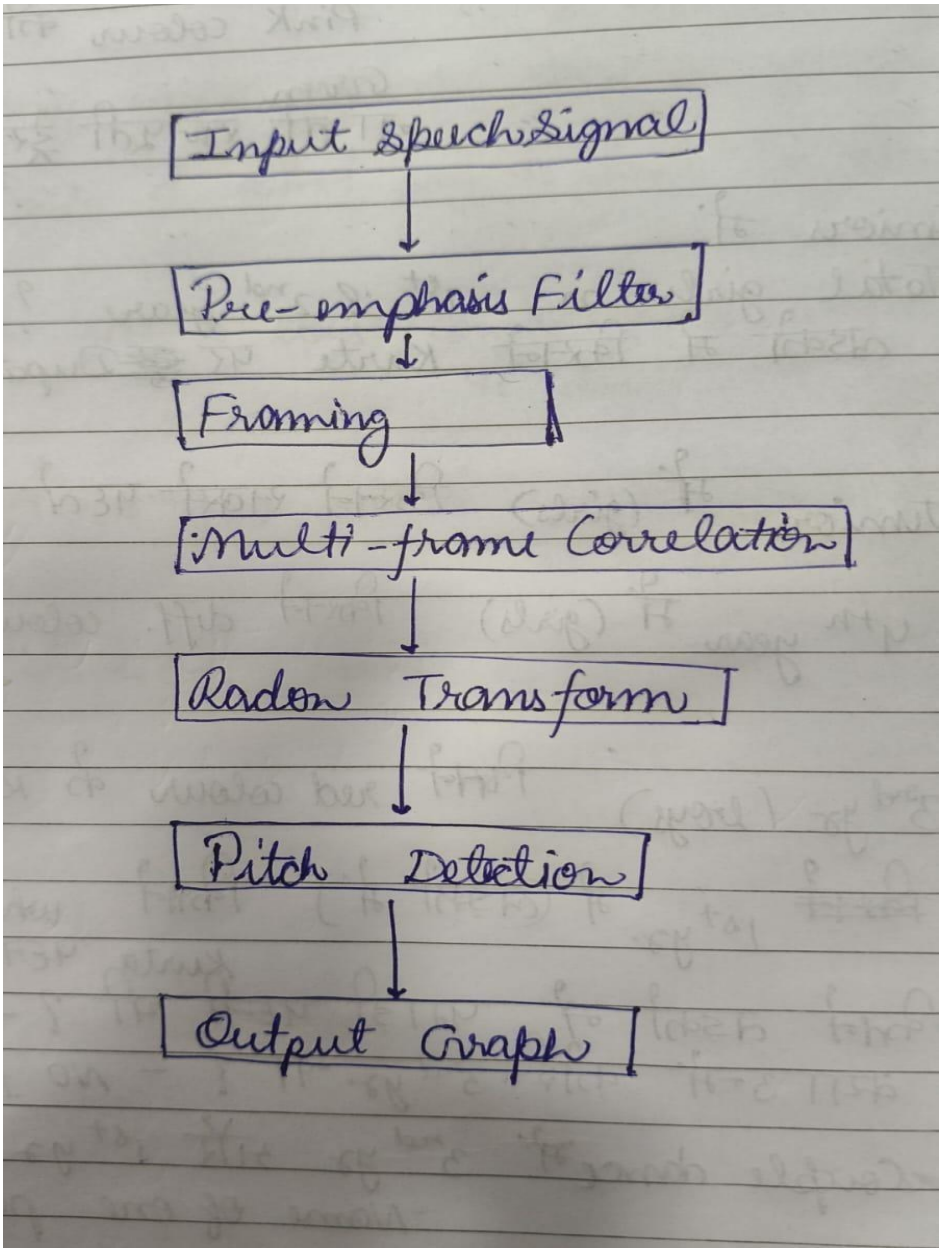
Implementation:

The MFCRT algorithm is implemented using Python. The file mfcrt.py contains the main logic of the algorithm. The evaluate.py file is used to run the code and generate results. Libraries like NumPy, SciPy, and Matplotlib are used in the implementation.

Results:

The output is shown as a graph of pitch versus time. The MFCRT method gives stable and accurate results even in noisy conditions. The graph shows that pitch detection is smooth and consistent.

Diagram:


Input speech Signal
Pre-emphasis Filter
Framing
Multi-frame Correlation
Radon Transform
Pitch Detection
Output Graph

Conclusion:

The MFCRT method is effective for pitch detection in noisy signals. It improves accuracy compared to traditional methods. The use of multi-frame correlation and Radon transform makes it reliable.

Limitations:

The algorithm is complex and requires more computation. It may not be suitable for real-time applications without optimization.

GitHub link:

GitHub Repository: https://github.com/khushbukataria55-commits/-Pitch-Detection-in-Noisy-Speech-Signals.git